

Product / Functional Requirements Document

No Food Waste — Food Rescue & Redistribution Platform

1. User Roles & Responsibilities

Role	Main Responsibility
Donor	Submit surplus-food requests, track status, view donor statistics, chat with volunteer.
Volunteer	Discover prioritized requests, claim tasks, track pickup & delivery, upload proof photos.
Admin	Manage volunteers/regions, review delivery proofs, monitor analytics and hunger hotspots.

2. Request Lifecycle State Machine

Standard flow: PENDING -> ACCEPTED -> IN_PROGRESS -> DELIVERED

Alternative flows: PENDING -> CANCELLED (by Donor); ACCEPTED -> PENDING (Volunteer rejection with reason)

3. End-to-End User Flow

- Donor registers and creates a food donation request specifying food type, quantity, servings, expiry, and pickup location.
- System prioritizes the request for nearby active volunteers using the distance and quantity scoring formula.
- A volunteer accepts the request atomically. Donor is notified immediately in real time.
- Volunteer navigates to pickup location, marks collection in progress, and coordinates via request-scoped chat.
- Volunteer delivers to beneficiaries/shelters and submits Delivery Proof (GPS coordinates, food photo, and delivery spot photo).
- Request transitions to DELIVERED; Donor, Volunteer, and Admin statistics and hunger hotspot data automatically update.